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Hair Braiding Device

Abstract

A hair braiding device for automatically braiding hair includes a housing which defines an orifice that extends into the housing. A motor is mounted in the housing and is operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction. A braiding mechanism is operably coupled to the output shaft of the motor and is mounted in the orifice of the housing. The braiding mechanism comprises a plurality of hair couplers which are configured to move along a figure-eight path when the motor is activated.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] I hereby claim the benefit under 35 U.S.C., Section 120 of U.S. application Ser. No.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC
OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT
INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to hair braiding devices and more particularly pertains to a new hair braiding device for automatically braiding hair.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art relates to hair braiding devices which twist hair to form a braid. Some such devices attach to ends of several tufts of hair via several hair couplers and cause the hair couplers to move in a figure-eight path to form a stronger braid than those in which the tufts of hair are merely twisting in a circular path. The prior art fails, however, to describe such an apparatus which is operable to move the hair couplers in either of a pair of opposing directions along the figure-eight path. Such a function would facilitate the formation of braids in a greater number of styles consisting of alternating braid patterns. The function would also make it possible to operate the device in reverse to untwist a braid. The devices of the prior art also fail to disclose a braiding mechanism which is removable from a motor of the device, which may allow for other attachments, including variations of the braiding mechanism, to be interchangeably placed in engagement with the motor.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a housing which defines an orifice that extends into the housing. A motor is mounted in the housing and is operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction. A braiding mechanism is operably coupled to the output shaft of the motor and is mounted in the orifice of the housing. The braiding mechanism comprises a plurality of hair couplers which are configured to move along a figure-eight path when the motor is activated. Each hair coupler is configured to attach to a tuft of hair.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will

become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. **1** is a perspective view of a hair braiding device according to an embodiment of the disclosure.

[0013] FIG. **2** is an exploded perspective view of an embodiment of the disclosure.

[0014] FIG. **3** is a front perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0015] FIG. **4** is a rear perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0016] FIG. **5** is a rear perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0017] FIG. **6** is a front perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0018] FIG. **7** is a top perspective detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0019] FIG. **8** is a rear detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0020] FIG. **9** is a front detail view of a braiding mechanism of an embodiment of the disclosure with a casing removed.

[0021] FIG. **10** is an exploded perspective detail view of a braiding mechanism of an embodiment of the disclosure.

[0022] FIG. **11** is a front detail view of a hair coupler of an embodiment of the disclosure.

[0023] FIG. **12** is a side detail view of a hair coupler of an embodiment of the disclosure.

[0024] FIG. **13** is a detail in-use view of an embodiment of the disclosure.

[0025] FIG. **14** is a block diagram of an embodiment of the disclosure.

[0026] FIG. **15** is an alternative embodiment of the disclosure with a charging station for charging a battery.

DETAILED DESCRIPTION OF THE INVENTION

[0027] With reference now to the drawings, and in particular to FIGS. **1** through **15** thereof, a new hair braiding device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0028] As best illustrated in FIGS. **1** through **15**, the hair braiding device **10** generally comprises a housing **12** which defines an orifice **14** that extends into the housing **12**. The housing **12** is elongated between a first end **16** and a second end **18**, and the orifice **14** is positioned on the first end **16** of the housing **12**. A motor **20** is mounted in the housing **12** and is operable to rotate an output shaft **22** of the motor **20** alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction.

[0029] A braiding mechanism **24** is operably coupled to the output shaft **22** of the motor **20**. The braiding mechanism **24** is mounted in the orifice **14** of the housing **12** and is removable from the housing **12** and the output shaft **22**. The braiding mechanism **24** comprises an input shaft **26**, a drivetrain **28**, a plurality of hair couplers **40**, and a track. The input shaft **26** is coupled to the output shaft **22** of the motor **20** and urges the drivetrain **28** to cause the hair couplers **40** to move in a figure-eight path as guided by the track. The input shaft **26** is axially aligned with the output shaft **22** wherein the output shaft **22** urges the input shaft **26** to rotate with respect to the housing **12** when the output shaft **22** rotates. The drivetrain **28** is operatively coupled to the input shaft **26** and includes a pair of output gears **30** which engage the hair couplers **40**.

[0030] The output gears **30** are equal in size to each other and have axes which are oriented parallel to each other. The output gears **30** are also enmeshed with each other such that the pair of output gears **30** rotate in opposite directions from each other. A front side **32** of each output gear **30** of the pair of output gears **30** faces outwardly away from the housing **12**. Each output gear **30** of the pair

of output gears **30** defines a plurality of cavities **34** in the front side **32** of the output gear **30**. The plurality of cavities **34** of each output gear **30** is radially arranged with respect to the output gear **30** and is open at a perimeter **36** of the output gear **30**.

[0031] Each hair coupler **40** of the plurality of hair couplers **40** is positioned in one of the cavities **34** of the plurality of cavities **34** of one of the output gears **30** of the pair of output gears **30**. Each hair coupler **40** comprises a rod **42**, a track engagement member **44**, and a hook **50**. The rod **42** is positioned in the one cavity **34** and protrudes out of the one cavity **34** beyond the front side **32** of the one output gear **30**. The track engagement member **44** is coupled to the rod **42** opposite the one output gear **30**, and the hook **50** is coupled to the track engagement member **44** opposite the rod **42**. The hook **50** is biased via a tension spring **51** toward the track engagement member **44** in a clamping fashion. The hooks **50** are attachable to hair **88** via clamping the hair between the hooks **50** and the respective track engagement members **44**, but the hair couplers **40** may, alternatively or in addition to the hooks **50**, eyelets, ties, or the like for such purpose.

[0032] The track engagement member **44** comprises a cam **46** and a pair of shoulders **48**. The cam **46** is elongated in an elongation direction which is perpendicular to the rod **42**. The cam **46** has an elliptical outer surface with a major axis that extends in the elongation direction, but other elongated shapes may be suitable. The shoulders **48** are positioned opposite each other across the cam **46**.

[0033] The track **52** is spaced forwardly of the pair of output gears **30** and engages the plurality of hair couplers **40**. The track **52** defines a groove **54** with a figure-eight shape **56**, which the hair couplers **40** are slidable along. The track **52** guides each hair coupler **40** of to move from the one cavity **34** of the one output gear **30** to another cavity **34** of another output gear **28** when the hair coupler **40** crosses through a central portion **58** of the figure-eight shape **56**. As depicted most clearly in FIG. **9**, the central portion **58** of the figure-eight shape **56** comprises a pair of intersecting straight portions **60**. The elongated shape of the cam **46**, which is positioned in the groove **54** of the track **52**, engages the walls of the groove **54** in one of these intersecting straight portions **60** to be guided across the other intersecting straight portion **60** rather than being allowed to move in a purely circular fashion. The shoulders **48** of the track **52** engagement portion of each hair coupler **40** are positioned opposite each other across the track **52** and act to retain the hair coupler **40** on the track **52**.

[0034] The Figures depict three hair couplers **40** corresponding to three cavities **34** in each output gear **30**. The cavities **34** of the output gears **30** are arranged such that each cavity **34** of one output gear **30** aligns with one of the cavities **34** of the other output gear **30** when these cavities **34** are positioned near the central portion **58** of the figure-eight shape **56**. In this arrangement, each hair coupler **40** may be passed alternately between pairs of aligning cavities **34**. Other quantities of hair couplers **40** and cavities **34** are possible and contemplated by the present invention. Other arrangements in which hair couplers **40** are passed through more than two cavities **34** may be used.

[0035] The drivetrain **28** depicted in the Figures shows other gears **38** which operatively couple the input shaft **26** of the drivetrain **28** to the output gears **30**. Myriad arrangements of these other gears **38** are possible, including various gear reducer arrangements for causing the output gears **30** to rotate at an output rotational speed which is slower than an input rotational speed of the output shaft **22** of the motor **20**. Arrangements in which the output rotational speed is faster than the input rotational speed may also be used. A casing **62** may be used to house or mount one or more gears of the drivetrain **28**. The casing **62** may also facilitate inserting and removing the drivetrain **28** into and out of the orifice **14** of the housing **12** respectively by having gripping surfaces ergonomically sized for grasping with a hand. The casing **62** may also facilitate positioning the drivetrain **28** in the orifice **14** by the nature of having a complementary outer surface **64** to a shape of the orifice **14**.

[0036] A connector **66** is coupled to the output shaft **22** of the motor **20** and the input shaft **26** of the braiding mechanism **24** and facilitates releasably connecting the braiding mechanism **24** to the motor **20**. The connector **66** comprises a first interlocking member **68** coupled to the output shaft

22 and a second interlocking member 72 coupled to the input shaft 26. The first interlocking member 68 has internal teeth 70 which mesh with external teeth 74 of the second interlocking member 72 to couple the output shaft 22 to the input shaft 26.

[0037] A grip 76 is coupled to and extends away from the housing 12. The grip 76 is oriented perpendicularly to the housing 12 and is positioned adjacent to a second end 18 of the housing 12. A processor 78 is operatively coupled to the motor 20. An activation switch 80 and a speed control 82 are operatively coupled to the processor 78. The activation switch 80 is actuatable to selectively activate the motor 20 to rotate the output shaft 22 in the first rotational direction or the second rotational direction. The speed control 82 is actuatable to select a speed of rotation of the motor 20. The activation switch 80 comprises a trigger positioned on the grip 76, and the speed control 82 is also mounted to the grip 76. A power cord 84 is electrically coupled to the processor 78 and is configured to electrically couple to an electrical power source. In other embodiments, a battery or other like power source may be housed in the housing 12 or the grip 76 and may be electrically coupled to the processor 78. As depicted in FIG. 15, a charging station 86 may electrically couple to a power source and to the battery to charge the battery in such embodiments.

[0038] In use, ends 92 of tufts 90 of hair 88 are attached respectively to the hooks 50 of the hair couplers 40 by clamping the tufts 90 between the hooks 50 and the track engagement members 44. The motor 20 is activated in a selected one of the first rotational direction and the second rotational direction and at a selected speed of rotation. The motor 20 causes the hair couplers 40 to move along the figure-eight path, braiding the tufts 90 of hair 88 together. After use of the device 10, the tufts 90 of hair 88 may be secured in the braided state by a hair tie or the like.

[0039] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0040] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A hair braiding device comprising: a housing, the housing defining an orifice which extends into the housing; a motor mounted in the housing, the motor being operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction; and a braiding mechanism being operably coupled to the output shaft of the motor, the braiding mechanism being mounted in the orifice of the housing, the braiding mechanism comprising a plurality of hair couplers being configured to move along a figure-eight path when the motor is activated, each hair coupler of the plurality of hair couplers being configured to attach to a tuft of hair.

2. The device of claim 1, wherein the braiding mechanism further comprises: an input shaft being coupled to and being axially aligned with the output shaft wherein the output shaft urges the input shaft to rotate with respect to the housing when the output shaft rotates; a drivetrain being

operatively coupled to the input shaft, the drivetrain including a pair of output gears, the pair of output gears being equal in size and having axes which are oriented parallel to each other, the pair of output gears being enmeshed with each other such that the pair of output gears rotate in opposite directions from each other, a front side of each output gear of the pair of output gears facing outwardly away from the housing, each output gear of the pair of output gears defining a plurality of cavities in the front side of the output gear, the plurality of cavities of each output gear of the pair of output gears being radially arranged with respect to the output gear, each cavity of the plurality of cavities being open at a perimeter of the output gear; wherein each hair coupler of the plurality of hair couplers is positioned in one of the cavities of the plurality of cavities of one of the output gears of the pair of output gears; and a track being spaced forwardly of the pair of output gears, the track engaging the plurality of hair couplers, the track being shaped to guide the plurality of hair couplers along the figure-eight path, the track guiding the plurality of hair couplers to be passed between the pair of output gears.

3. The device of claim 2, wherein each hair coupler further comprises: a rod positioned in the one cavity and protruding out of the one cavity beyond the front side of the one output gear; a track engagement member being coupled to the rod opposite the one output gear, the track engagement member comprising: a cam, the cam being elongated in an elongation direction which is perpendicular to the rod; and a pair of shoulders being coupled to the cam, the pair of shoulders being positioned opposite each other across the cam; wherein the track defines a groove having a figure-eight shape, the plurality of hair couplers being slidable along the groove, the track guiding each hair coupler of the plurality of hair couplers to move from the one cavity of the one output gear to another cavity of another gear when the hair coupler crosses through a central portion of the figure-eight shape, the cam of the track engagement member of each hair coupler of the plurality of hair couplers being positioned in the groove, the pair of shoulders of the track engagement member of each hair coupler of the plurality of hair couplers being positioned opposite each other across the track.

4. The device of claim 3, wherein the cam has an elliptical outer surface with a major axis extending in the elongation direction.

5. The device of claim 2, wherein the braiding mechanism is removable from the housing and the output shaft, further comprising a connector being coupled to the output shaft of the motor and the input shaft of the braiding mechanism, the connector comprising a first interlocking member coupled to the output shaft and a second interlocking member coupled to the input shaft, the first interlocking member and the second interlocking member mating with each other to couple the output shaft to the input shaft.

6. The device of claim 5, wherein the first interlocking member having internal teeth which mesh with external teeth of the second interlocking member to couple the output shaft to the input shaft.

7. The device of claim 1, wherein the housing is elongated between a first end and a second end, the orifice being positioned on the first end of the housing, further comprising a grip being coupled to and extending away from the housing, the grip being oriented perpendicularly to the housing, the grip being positioned adjacent to a second end of the housing.

8. The device of claim 1, further comprising: a processor being operatively coupled to the motor; an activation switch being operatively coupled to the processor, the activation switch being actuatable to selectively activate the motor to rotate the output shaft in the first rotational direction or the second rotational direction; and a speed control being operatively coupled to the processor, the speed control being actuatable to select a speed of rotation of the motor.

9. The device of claim 8, further comprising a power cord being electrically coupled to the processor, the power cord being configured to electrically couple to an electrical power source.

10. The device of claim 1, wherein each hair coupler of said plurality of hair couplers comprises a hook which is coupled to the track engagement member opposite the rod.

11. The device of claim 10, wherein the hook is biased toward the track engagement member in a

clamping fashion.

12. A hair braiding device comprising: a housing, the housing defining an orifice which extends into the housing, the housing being elongated between a first end and a second end, the orifice being positioned on the first end of the housing; a motor mounted in the housing, the motor being operable to rotate an output shaft of the motor alternately in a first rotational direction and a second rotational direction opposite from the first rotational direction; a braiding mechanism being operably coupled to the output shaft of the motor, the braiding mechanism being mounted in the orifice of the housing, the braiding mechanism being removable from the housing and the output shaft, the braiding mechanism comprising: an input shaft being coupled to and being axially aligned with the output shaft wherein the output shaft urges the input shaft to rotate with respect to the housing when the output shaft rotates; a drivetrain being operatively coupled to the input shaft, the drivetrain including a pair of output gears, the pair of output gears being equal in size and having axes which are oriented parallel to each other, the pair of output gears being enmeshed with each other such that the pair of output gears rotate in opposite directions from each other, a front side of each output gear of the pair of output gears facing outwardly away from the housing, each output gear of the pair of output gears defining a plurality of cavities in the front side of the output gear, the plurality of cavities of each output gear of the pair of output gears being radially arranged with respect to the output gear, each cavity of the plurality of cavities being open at a perimeter of the output gear; a plurality of hair couplers, each hair coupler of the plurality of hair couplers being positioned in one of the cavities of the plurality of cavities of one of the output gears of the pair of output gears, each hair coupler comprising: a rod positioned in the one cavity and protruding out of the one cavity beyond the front side of the one output gear; a track engagement member being coupled to the rod opposite the one output gear, the track engagement member comprising: a cam, the cam being elongated in an elongation direction which is perpendicular to the rod, the cam having an elliptical outer surface with a major axis extending in the elongation direction; and a pair of shoulders being coupled to the cam, the pair of shoulders being positioned opposite each other across the cam; and a hook being coupled to the track engagement member opposite the rod, the hook being biased toward the track engagement member in a clamping fashion; and a track being spaced forwardly of the pair of output gears, the track engaging the plurality of hair couplers, the track defining a groove having a figure-eight shape, the plurality of hair couplers being slidable along the groove, the track guiding each hair coupler of the plurality of hair couplers to move from the one cavity of the one output gear to another cavity of another gear when the hair coupler crosses through a central portion of the figure-eight shape, the cam of the track engagement member of each hair coupler of the plurality of hair couplers being positioned in the groove, the pair of shoulders of the track engagement member of each hair coupler of the plurality of hair couplers being positioned opposite each other across the track; a connector being coupled to the output shaft of the motor and the input shaft of the braiding mechanism, the connector comprising a first interlocking member coupled to the output shaft and a second interlocking member coupled to the input shaft, the first interlocking member having internal teeth which mesh with external teeth of the second interlocking member to couple the output shaft to the input shaft; a grip being coupled to and extending away from the housing, the grip being oriented perpendicularly to the housing, the grip being positioned adjacent to a second end of the housing; a processor being operatively coupled to the motor; an activation switch being operatively coupled to the processor, the activation switch being actuatable to selectively activate the motor to rotate the output shaft in the first rotational direction or the second rotational direction, the activation switch comprising a trigger positioned on the grip; a speed control being operatively coupled to the processor, the speed control being actuatable to select a speed of rotation of the motor, the speed control being mounted to the grip; and a power cord being electrically coupled to the processor, the power cord being configured to electrically couple to an electrical power source.
